

Volume Estimator

Update: Text-Based Selection

Type what to measure — the app finds it, outlines it, and estimates only that region.

NEW

Depth Anything V2

LangSAM

June 14, 2026

Why we added it

The problem

- ▶ Earlier the app measured the *whole frame* only — no way to isolate one object in a cluttered scene.
- ▶ Manual masks (clicks / boxes) are slow and need a trained operator.

The goal

- ▶ Let *anyone* pick the target in plain English.
- ▶ Measure only the object of interest, not the background.
- ▶ Keep the accurate depth pipeline that already worked best.

What changed

From

Full-frame only



To

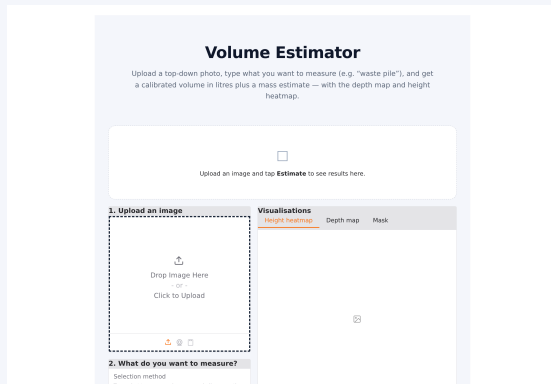
Type “waste pile” → measured

What's new: text-based selection

Text selection is now the **default** mode:

1. Upload a top-down photo.
2. **Type** the object — e.g. “waste pile”, “cardboard boxes”.
3. Press **Estimate**.

No clicking on the image, no drawing boxes — the prompt box is front-and-centre, with larger, higher-contrast UI.



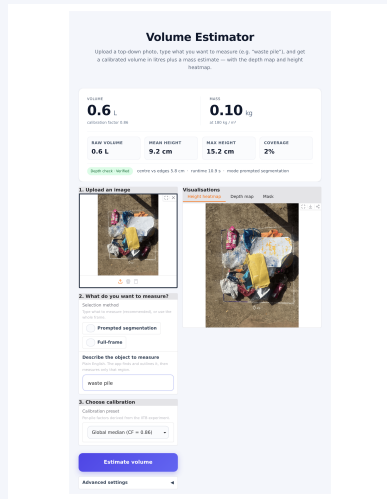
The redesigned portal — prompt-driven UI.

Under the hood

Pipeline

1. **Text** → **mask**: LangSAM (GroundingDINO + SAM 2.1) turns the prompt into a pixel mask.
2. **Depth**: Depth Anything V2 (Metric Indoor) — the model that gave our best results.
3. **Height + volume**: floor-plane fit, calibrated scale, integrate height inside the mask only.

Output: calibrated volume (L), mass (kg), plus depth map, height heatmap and the selection mask.



Live result: prompt “waste pile” measured end-to-end.

Summary & next steps

Shipped

- ▶ Prompt-driven (text-based) selection — now the default mode.
- ▶ Same best-performing depth model (Depth Anything V2).
- ▶ Cleaner, larger, higher-contrast UI; full-frame mode still available.

Next

- ▶ Validate prompt segmentation accuracy across the IITB dataset.
- ▶ Multi-object prompts and per-object volume breakdown.
- ▶ Default the visual to the “Mask” tab so users see the selection.

Try it

running locally at <http://localhost:7862>